

A Dual-Model Portfolio for Identity-Document Fraud Detection: Resolution as the Universal Lever

The FREUID Challenge 2026 (IJCAI-ECAI) — Technical Report

Team: EliasTSJ + Sekinal · July 2026 · Public LB: **0.02514**

1. Introduction

The FREUID Challenge asks for a fraud score per identity-document image across five African countries, with a public leaderboard computed on ~5% of the test set and a final private evaluation announced to contain (a) two document types / countries absent from training, and (b) an emphasis on real print-and-capture examples that suppress fragile digital artifacts.

This design creates a fundamental tension: the strategies that maximize the public score (exploiting fragile pixel-level traces of digital manipulation) are exactly the ones announced to be penalized on the private set. Our solution embraces the tension instead of resolving it: we submit a **two-model portfolio** — a *public specialist* and a **capture-robust generalist** — trained identically except for two flags, and we validate every robustness decision with **Leave-One-Country-Out (LOCO)** protocols rather than the public leaderboard.

Our central empirical finding is that ****input resolution dominates every other design choice****, improving in-distribution and out-of-distribution performance simultaneously and by an order of magnitude: public LB $0.123 \rightarrow 0.025$ and unseen-country FREUID $0.40 \rightarrow 0.01\text{--}0.03$ when moving from 768 px to 1024 px (near-native). The forensic traces of document manipulation live at the pixel scale, and they transfer across countries — document templates do not need to.

2. Data

2.1 Competition data (used by the final models)

The final models train **exclusively on the official FREUID training set**:

69,352 images, five types (EGYPT/DL, GUINEA/DL, BENIN/DL, MOZAMBIQUE/DL, MAURITIUS/ID), fraud rate 42.8%, native resolutions up to 1585×1000 (median width 1000). Only 20 training images are non-digital (captured); we reserve them as a "canary" probe (§4.3). No pseudo-labels, no external data, no public-test images are used in training the final submissions.

2.2 External sources (explored, measured, and rejected)

We evaluated four external augmentation strategies. Each was rejected on measurement — a result we consider a key finding (§5.4):

External source	License	Result
IDNet-2025 digital (~88k EU documents)	CC-BY-4.0	No LOCO gain; >30k rows add noise on the public LB
IDNet-2025 scanned (print-captured)	CC-BY-4.0	Destroys real-capture detection (canary AUC 0.50)
Synthetic GenAI fraud, 7.5k images we generated (InsightFace inswapper face-swap, SDXL-inpainting face/text edits, classic splices over FREUID genuines)	model-specific	Teaches "unknown country $\Rightarrow$ fraud": held-out-country FREUID degrades to 1.0 (\$5.4)
FantasyID (bona-fide capture set)	research	Neutral; dropped with the rest
Public-test pseudo-labels (transductive self-training)	—	Helps only while the teacher is stronger than the student; harmful at our final level (\$5.5)

3. Method

3.1 Architecture

Both final models are a `timm EfficientNetV2-M` (`tf_efficientnetv2_m.in21k_ft_in1k`) with a small MLP head (LayerNorm $\rightarrow$ Dropout $\rightarrow$ Linear(1280 $\rightarrow$ 512) $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear(512 $\rightarrow$ 1)), trained with BCE at **1024 $\times$ 1024** input resolution, batch 12, AdamW, lr 1e-4, on a single NVIDIA L40S (48 GB).

3.2 The two training regimes

	■ Public specialist	■ Capture-robust generalist
Capture-simulating augmentation	off	heavy
Epochs	3	1 (early stop)
Seeds	1 (seed 4)	3, probability-averaged
Horizontal-flip TTA	yes	yes

Capture-simulating augmentation (the single most important robustness component) re-degrades every training image randomly: JPEG re-encoding at quality 22–90, downscale–upscale (0.3–1.0 $\times$), Gaussian blur, moiré synthesis, perspective warp, and photometric jitter. The public specialist disables it to preserve the fragile digital artifacts the public test rewards; the robust model relies on it to learn manipulation evidence that survives print-and-capture.

Early stopping is the second robustness lever: with capture-aug on, held-out-country FREUID degrades monotonically with training (0.025 at epoch 0 $\rightarrow$ 0.097 at epoch 2 on Mozambique-LOCO) as the network drifts from general manipulation evidence toward memorizable digital artifacts. Without capture-aug at 1024 px the effect reverses (0.151 $\rightarrow$ 0.033), indicating the two regimes converge to different solution families.

3.3 What we tried that did not survive (mid-competition lineage)

Before the resolution finding, our best public model was a ConvNeXtV2-L pipeline: domain-adaptive masked-autoencoder pretraining on FREUID images

(public 0.172 → 0.156), then 512 px fine-tuning (0.153). Approaches that failed outright: frozen foundation-model probes (DINOv2/CLIP, 0.24–0.37), soft-AUC ranking loss (0.37), REINFORCE fine-tuning (0.175), SRM noise-residual stream (1.0), soft pseudo-labels (0.35), DANN domain-adversarial training (LOCO 0.92 vs 0.40 baseline), and every cross-architecture ensemble (consistent dilution). The final EffNetV2-M @1024 recipe supersedes the entire lineage on both axes.

4. Validation protocols (the compass)

The public LB is 5% of the test data and rewards transductive overfitting, so all robustness decisions used local protocols:

4.1 Leave-One-Country-Out (LOCO)

Hold out one full country/type from training; evaluate the competition metric on it. We ran LOCO on three countries (Mozambique, Benin, Mauritius) to avoid tuning to a single held-out choice. Historical baselines: 0.40 (Mozambique) and 0.39 (Benin) — the private-test scenario was genuinely hard for every recipe before the resolution finding.

4.2 Seed-noise calibration

Identical 1-epoch runs differ by up to ± 0.05 LOCO FREUID; we treated sub-0.05 differences as inconclusive and re-ran candidates, and we used probability-averaged seed ensembles (which beat the best individual seed: 0.131 vs 0.136) for the final robust model.

4.3 Capture canary

The 20 captured training images (held out of LOCO training runs that exclude Mauritius, their majority country) probe print-and-capture robustness: capture-aug models reach AUC 0.917 with clean separation; no-capture-aug models degrade to 0.726 and over-flag captured genuines — the measured cost of public-LB specialization, and the reason the portfolio has two models.

5. Results

5.1 Final models

Model	Public LB	LOCO Mozambique	LOCO Benin	LOCO Mauritius	Canary AUC
■ Public specialist @1024	0.02514	0.033	0.0101	0.0069	0.726
■ Robust ensemble @1024	0.11925	0.025 (ep 0)	—	0.0156	0.917

5.2 The resolution ladder

Public LB (same recipe, resolution only): 384 → ~0.19 · 512 → 0.153 · 768 → 0.123 · 896 → 0.034 · **1024** → **0.025**. Unseen-country (LOCO

Mozambique): 384 → 0.40 · 768 → 0.15–0.22 · **1024** → **0.03**. Native images are up to 1585×1000: at 768 px, training destroys the resampling seams, JPEG-grid mismatches, and interpolation traces that identify manipulation; at 1024 px ($\approx$ native height) they reach the network intact. Returns were *increasing*, not diminishing — evidence the decisive signal lives at the finest scale. Verified as legitimate out-of-sample performance: no public-test data in training, bimodal score distribution, inter-resolution agreement $r = 0.91$, and LOCO transfer (this section).

5.3 Progression (public LB)

0.333 baseline → 0.291 leakage-safe splits → 0.207 aug+focal → 0.187 (+30k IDNet) → 0.178 (+TTA) → 0.172 (pseudo-labels) → 0.156 (domain-MAE) → 0.153 (512 px) → 0.123 (EffNetV2-M 768 px) → 0.034 (896 px) → **0.025 (1024 px)**.

5.4 External data poisoning (negative result we consider a contribution)

Synthetic GenAI fraud generated from FREUID genuines collapses LOCO to 1.0: the synthetic edits include held-out-country genuines re-labeled as fraud, so the network learns "unfamiliar country $\Rightarrow$ fraud" — the worst possible prior for a private set of unseen countries. The mechanism is general: *synthetic attack data built from in-domain genuines silently leaks country identity into the fraud label.* Similarly, IDNet-scanned (print-captured European documents) reduces the capture canary to chance (AUC 0.50): off-domain capture statistics displace, rather than teach, the target capture distribution.

5.5 Pseudo-labels and self-training limits

Public-test pseudo-labels helped while the teacher outperformed the student (0.187 → 0.172 era). At the final level they consistently hurt: a 0.153 teacher drags a 0.123-recipe student to 0.175; even self-training with the best available teacher (0.123) lands at 0.137. With 99.6% of pseudo-labels above the confidence threshold, pseudo-labeling is effectively distillation of the teacher's errors onto the test distribution.

6. Inference

Single pass over the input directory (`infer.py`): each decoded batch is scored by every checkpoint and TTA view before the next batch loads, so decode cost is amortized. bf16 autocast, channels-last, batch 32.

Runtime (measured, NVIDIA L40S 48 GB, decoding native-resolution JPEGs): the 3-seed + hflip-TTA robust ensemble (6 forwards/image at 1024 px) runs at ≈ 19 img/s → a 135k-image hidden test set completes in ≈ 2 h, well inside the 6-hour single-A100 budget (the A100-80GB is $\geq$ the L40S on bf16 throughput). The single-model public specialist is $\sim 3\times$ faster (≈ 40 min).

7. Reproducibility

- **Code:** public repository (MIT license), frozen commit; layout and exact training commands in `README.md`. Total training compute for the final models: ~3 GPU-hours (1× L40S).
- **Weights:** 4 checkpoints (~216 MB each) attached as a GitHub Release (`final-models`).
- **Docker:** `docker/Dockerfile` implements the sandbox contract — reads `/data` (read-only, flat images), writes `/submissions/submission.csv`, runs with `--network none` (weights and code embedded; no downloads). Default entrypoint produces the robust-ensemble submission; `FREUID_MODEL=public` switches to the public specialist.
- **Hardware for verification:** any single NVIDIA GPU ≥ 24 GB (A100 recommended); CPU fallback works but is slow.
- **Determinism note:** training used unfixed seeds where noted; the released checkpoints are the exact artifacts behind the final submissions, so verification is inference-only and deterministic up to bf16 nondeterminism (rank correlation ≥ 0.999 across reruns).

8. Acknowledgements

Built with PyTorch, torchvision, and timm (Apache-2.0). We thank the organizers (Microblink Fraud Lab) for a competition design that rewards genuine robustness — and for the forum guidance that shaped the portfolio strategy.